

## [Sophia Exclusive] Grade 5A Workbook Error Remediation & Ch6 Review Mastery Worksheet

Student Name:  
Sophia

Suggested Time: 45 ~ Total Points:  
50 Mins 100

Target:

5A Targeted Remediation + Ch6 Review Blind Spots

### 💡 Pre-Test Pitfall Avoidance Rules:

① Circle keywords & units before writing; pair every quantity with its correct unit! ② Estimate before calculating; never invert operations! ③ Keep algebraic variables and constant numbers separate; always divide by 2 for triangle area!

### Part 1: Conceptual Precision & Number Sense (Fill-in-the-Blanks & Comparison)

5 Questions, 20 Points

#### 1. [Target ERR-SOP-13] Quotient Invariance & Decimal Point Scaling (4 pts)

Rule: When divisor stays constant, scaling the dividend scales the quotient equally. Small dividend / big divisor = quotient  $< 1$ !

Given  $48 \div 16 = 3$ , write down the quotients directly using the rules of division:

(1)  $4.8 \div 16 =$  \_\_\_\_\_ (2)  $0.48 \div 16 =$  \_\_\_\_\_ (3)  $0.048 \div 16 =$  \_\_\_\_\_

(4)  $480 \div 16 =$  \_\_\_\_\_

#### 2. [Target ERR-SOP-15] Comparing Operations with Decimals Less Than 1 (4 pts)

Rule: Multiplying by a number  $< 1$  shrinks the value; dividing by a number  $< 1$  enlarges the value!

Fill in each  $\square$  with " $>$ ", " $<$ ", or " $=$ " without detailed calculation:

(1)  $6.4 \times 0.8 \square 6.4 \div 0.8$  (2)  $12.5 \div 0.5 \square 12.5 \times 0.5$  (3)  $0.95 \times 1.02 \square 0.95 \div 1.02$

(4)  $8.4 \div 1 \square 8.4 \times 1$

#### 3. [Target ERR-SOP-18] Recurring Decimals & Remainder Cycles (4 pts)

Rule: Expand recurring digits to compare; use division remainder to pinpoint the Nth recurring digit!

(1) Compare:  $0.714714 \square 0.\dot{7}1\dot{4}$  (Hint: expand and compare the 7th decimal place).

(2) The fraction  $\frac{1}{7} = 0.\dot{1}4285\dot{7}$  has a repeating period of \_\_\_\_\_ digits. The 100<sup>th</sup> digit after the decimal point is \_\_\_\_\_.

#### 4. [Target ERR-SOP-24] Generality of Algebraic Variables (4 pts)

Rule: Variables represent any value; never substitute specific constants into general formulas!

A square has side length  $a$  cm:

(1) Its perimeter is \_\_\_\_\_ cm; (2) Its area is \_\_\_\_\_  $\text{cm}^2$ ; (3) When  $a = 6$ , its area is \_\_\_\_\_  $\text{cm}^2$

### 5. [Target ERR-SOP-27] Multiplicative Comparison Sentence Structures (4 pts)

Rule: "How many times B is A?" means  $A \div B$ ; identify the reference base carefully!

(1) "How many times 15 is 45?" Equation: \_\_\_\_\_, Result: \_\_\_\_\_

(2) "A car traveled 240 km, which is 3 times the distance a bike traveled." The bike's distance is \_\_\_\_\_ km.

## Part 2: Algebraic Simplification & Sign Rules (True/False & Evaluation)

3 Questions, 15 Points

### 6. [Target ERR-SOP-21] Combining Like Terms vs. Constants (5 pts)

Rule: Only like terms with identical variables can be combined; never add a constant into variable coefficients!

True or False: The algebraic expression  $4x + 7 + 2x$  simplifies to  $13x$ . (\_\_\_\_\_)

Write the correct simplification:  $4x + 7 + 2x =$

Explain why:

### 7. [Target ERR-SOP-22] Additive Inverses & Sign Properties (5 pts)

Rule: Signs stay with their terms;  $+y$  and  $-y$  cancel out to 0, never to  $-2y$ !

True or False: The identity  $x + x - y + y = 2x - 2y$  is always correct. (\_\_\_\_\_)

Write the standard step-by-step simplification:  $x + x - y + y =$

### 8. [Target ERR-SOP-24] "Simplify First, Then Substitute" Discipline (5 pts)

Rule: Always simplify expressions completely before evaluating numerical values!

Given  $y = 3.5$ , evaluate the algebraic expression  $15y - 7y + 20$  (**Direct substitution without simplifying is strictly prohibited**).

Step 1 (Combine like terms):  $15y - 7y + 20 =$

Step 2 (Substitute value): When  $y = 3.5$ , Value =

**Part 3: Rigorous Column Arithmetic & Mixed Operations****3 Questions, 20 Points****9. [Target ERR-SOP-14] Column Division with Decimal Divisors & Quotient Alignment (6 pts)**

Rule: Shift divisor decimal point to make it an integer; shift dividend identically; align quotient's decimal point!

Calculate:  $197.6 \div 5.2$ (1) **Pre-calculation estimate:** Round 197.6 to \_\_\_\_\_, 5.2 to \_\_\_\_\_, estimated quotient  $\approx$  \_\_\_\_\_.(2) **Column division** (clearly show shifted decimal points):

$$5.2 \overline{) 197.6}$$

Verification Check:

Is quotient an integer or decimal?

Final result =

**10. [Target ERR-SOP-20] Triangle Area Formula & Rigor (6 pts)**

Rule: Triangles and trapezoids must always be divided by 2; omitting division by 2 doubles the area!

A right triangle has legs of base  $b = 0.36$  m and height  $h = 0.25$  m.(1) Write the triangle area formula:  $\text{Area} = \underline{\hspace{2cm}}$ 

(2) Calculate the area of this triangle (show step-by-step working and units):

Working:

Answer:

**11. [Target ERR-SOP-26] Order of Operations with Decimals (8 pts)**

Rule: Parentheses first, then multiplication/division from left to right; count decimal places carefully!

(1)  $0.4 \times (3.2 - 0.8) \div 1.2$

(2)  $7.4 \times 1.3 - 4.68 \div 0.9$

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**Part 4: Word Problems & Keyword Modeling****4 Questions, 25 Points****12. [Target ERR-SOP-16] Two-Step Equal Sharing & Deducting Remainder (6 pts)**

Rule: When there is a remainder, subtract the remaining amount first before dividing into equal groups!

A bakery made a batch of cookies weighing 9.2 kg in total. The baker distributed them equally into 4 gift boxes, leaving 0.4 kg of cookies on the counter. How many kilograms of cookies were placed in each gift box?

Guided thinking: Total cookies actually packed into the boxes = \_\_\_\_\_ kg.

Equation &amp; Working:

Answer:

### 13. [Target ERR-SOP-17] Multi-Variable Matching & Unit Alignment (6 pts)

Rule: Match unit price with corresponding quantity; never multiply kilograms by kilograms!

At a supermarket, apples cost \$6.8 per kg and bananas cost \$4.5 per kg. Ms. Wang bought 15 kg of apples and 24 kg of bananas for the school cafeteria. How much did she pay in total?

Equation & Working:

Answer:

### 14. [Target ERR-SOP-23] Average Score Total Delta Model (6 pts)

Rule: Total = Average  $\times$  Number of Games; the difference in totals gives the score of the extra games!

The school girls' basketball team averaged 42 points in their first 6 tournament games. They then played 2 more games, raising their overall average across all 8 games to 45 points.

(1) How many total points did the team score in the last 2 games combined? (2) What was their average score in the last 2 games?

Step-by-step Working:

Answer: (1) Total points in last 2 games: \_\_\_\_\_ pts; (2) Average in last 2 games: \_\_\_\_\_ pts.

### 15. [Target ERR-SOP-19 & ERR-SOP-27] Sharing Pitfall & Fraction of the Whole (7 pts)

Rule: "Herself and 3 friends" =  $1 + 3 = 4$  people! Fraction of whole uses 1 as whole, not number of slices!

"Emma bought a large pizza for lunch and cut it into 8 equal slices. She wants to share the pizza evenly between **herself** and **three** of her best friends."

(1) How many people are sharing the pizza altogether? \_\_\_\_\_ people.

(2) What **fraction** of the whole pizza will each person receive? \_\_\_\_\_ (fraction) = \_\_\_\_\_ (decimal).

(3) How many **slices** of pizza will each person get? \_\_\_\_\_ slices.

**16. [Target 5A P44 ERR-SOP-25] Bar Model Total Length Conservation Equation (5 pts)**

Rule: Equal ribbons mean equal total length; solve for  $x$  first, then equate the total length to find  $y$ !

The bar models below show three ribbons of identical total length: Ribbon 1 is  $3x$ , Ribbon 2 is  $x + 8$ , and Ribbon 3 is  $4y$ .

[Tier 1] |---- x ----|---- x ----|---- x ----| (Total  $3x$ )  
 [Tier 2] |---- x ----|----- 8 -----| (Total  $x + 8$ )  
 [Tier 3] |-- y --|-- y --|-- y --|-- y --| (Total  $4y$ )

(1) Since Ribbon 1 and Ribbon 2 have equal length, set up an equation in terms of  $x$  and solve:

Equation: \_\_\_\_\_  $\Rightarrow x =$  \_\_\_\_\_

(2) Find the total length of each ribbon, set up an equation for  $y$ , and solve:

Total length = \_\_\_\_\_ dm; Equation for  $y$ : \_\_\_\_\_  $\Rightarrow y =$  \_\_\_\_\_

**17. [Target 5A P47 Work Collaboration] Road Paving Collaboration Problem (5 pts)**

Engineering teams A and B work together to pave a highway measuring 125 km in total. Team A paves 3.5 km per day, and Team B paves 4.5 km per day. Team A worked alone for the first 6 days due to scheduling, after which Team B joined to work together. How many days will it take to finish the remaining highway after Team B joins?

Working:

Answer:

**18. [Target 5A P49 Encounter Travel] Traveling Towards Each Other Equation (5 pts)**

City A and City B are 540 km apart. A coach and a truck depart simultaneously from City A and City B respectively, traveling towards each other. They meet after 4.5 hours. The average speed of the coach is 64 km/h. Let the truck's speed be  $x$  km/h. **Set up an algebraic equation and solve for  $x$ .**

Let the truck's average speed be  $x$  km/h.

Equation:

Answer:

**19. [Target 5A P51 Boat Rental Surplus & Deficit] Surplus and Shortage Algebraic Model (5 pts)**

SUIS organized Grade 5 teachers and students for a boating activity in a park. If each boat holds 4 people, 6 people are left without a boat. If each boat takes 2 more people (i.e. 6 people per boat), exactly 1 boat is completely empty. How many boats were originally rented? How many total teachers and students participated?

Let the number of rented boats be  $x$ .

Total people in Scheme 1: \_\_\_\_\_; Total people in Scheme 2: \_\_\_\_\_

Equation & Solution:

Answer:

## [Answer Key & Marking Rubric] (For Teacher Reference)

**Part 1 (20 pts):** 1. (1) 0.3 (2) 0.03 (3) 0.003 (4) 30 (1 pt each); 2. (1) < (2) > (3) < (4) = (1 pt each); 3. (1) < (2 pts) (2) 6, 8 (1 pt each); 4. (1)  $4a$  (2)  $a^2$  (3) 36 (1+1+2 pts); 5. (1)  $45 \div 15$ , 3 (2 pts) (2) 80 (2 pts).

**Part 2 (15 pts):** 6. False (2 pts); Simplification:  $6x + 7$  (2 pts); Reason: Only like terms can combine coefficients, constant 7 cannot merge with  $x$  (1 pt). 7. False (2 pts); Steps:  $(x + x) + (-y + y) = 2x + 0 = 2x$  (3 pts). 8. Simplify:  $8y + 20$  (2 pts); Evaluate:  $8 \times 3.5 + 20 = 28 + 20 = 48$  (3 pts).

**Part 3 (20 pts):** 9. (1) Estimate:  $200 \div 5 = 40$  (2 pts); (2) Column division: quotient is integer 38 (4 pts; deduct 3 pts if misplaced decimal point). 10. (1)  $S = \frac{1}{2}bh$  (2 pts); (2)  $0.36 \times 0.25 \div 2 = 0.045 \text{ m}^2$  (4 pts; deduct 3 pts if forgot to divide by 2). 11. (1)  $0.4 \times 2.4 \div 1.2 = 0.96 \div 1.2 = 0.8$  (4 pts); (2)  $9.62 - 5.2 = 4.42$  (4 pts).

**Part 4 (25 pts):** 12. Guide: 8.8 kg (1 pt);  $(9.2 - 0.4) \div 4 = 2.2 \text{ kg}$  (4 pts); Statement (1 pt). 13.  $6.8 \times 15 + 4.5 \times 24 = 102 + 108 = \$210$  (6 pts). 14. (1)  $45 \times 8 - 42 \times 6 = 360 - 252 = 108 \text{ pts}$  (4 pts); (2)  $108 \div 2 = 54 \text{ pts}$  (2 pts). 15. (1) 4 people (2 pts); (2)  $\frac{1}{4}$ , 0.25 (2 pts); (3) 2 slices (3 pts).

**Part 5 (20 pts):** 16. (1)  $3x = x + 8 \implies 2x = 8 \implies x = 4$  (2 pts); (2) Total length 12 dm (1 pt),  $4y = 12 \implies y = 3$  (2 pts). 17. Remaining:  $125 - 3.5 \times 6 = 104 \text{ km}$ ;  $104 \div (3.5 + 4.5) = 13 \text{ days}$  (5 pts). 18. Equation:  $(64 + x) \times 4.5 = 540 \implies 64 + x = 120 \implies x = 56 \text{ km/h}$  (5 pts). 19. Expressions:  $4x + 6$  and  $6(x - 1)$  (2 pts); Equation:  $4x + 6 = 6(x - 1) \implies 2x = 12 \implies x = 6 \text{ boats}$  (2 pts); Total people:  $4 \times 6 + 6 = 30 \text{ people}$  (1 pt).